



Constellation®

LaSalle County Station

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Marseilles, IL 61341
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RA24-016

10 CFR 50.73

May 17, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2024-002-00, Reactor Protection System Half Scram
due to Motor Generator Set Electric Power Monitoring Assembly Output Breaker
Trip

In accordance with 10 CFR 50.73(a)(2)(iv)(A), Constellation Energy Generation, LLC (CEG)
is submitting Licensee Event Report (LER) Number 2024-002-00 for LaSalle County
Station, Unit 1.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Ms. Laura Ekern,
Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

Christopher J. Smith
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report 2024-002-00

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00373

3. Page

1 OF 3

4. Title

Reactor Protection System Half Scram due to Motor Generator Set Electric Power Monitoring Assembly Output Breaker Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	25	2024	2024	- 002 -	00	05	17	2024	NA	☐ 052	NA

9. Operating Mode

1

10. Power Level

100 percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Joseph Reda, Operations Director

Phone Number (Include area code)

(815) 415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	JE	CBD	G080	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 25, 2024, LaSalle Unit 1 was operating at 100 percent power, when the 1B reactor protection system (RPS) motor-generator (MG) set electric power monitoring assembly (EPMA) output breaker tripped due to an underfrequency condition, which resulted in the loss of the 1B RPS bus. The loss of power to this bus created a half-scram condition and initiated closure of isolation valves in multiple systems. The condition was an invalid actuation because it was not the result of actual plant conditions and not an intentional manual action. Actuators and isolations that occurred performed as designed.

The cause of the MG set EPMA output breaker trip was the EPMA trip card for the output breaker being out of range high for the underfrequency setpoint which caused the trip on underfrequency. The loss of the 1B RPS bus resulted in general containment isolation signals that affected containment isolation valves in more than one system while the unit was critical. Therefore, the condition is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

LaSalle County Station, Unit 1

☒ 050☐ 052**2. DOCKET NUMBER**

00373

3. LER NUMBER

YEAR

2024

SEQUENTIAL

NUMBER

- 002

REV

NO.

- 00

NARRATIVE**Plant and System Identification**

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was "B" reactor protection system (RPS) logic and associate isolation functions. The affected component is the 1B electric power monitoring assembly (EPMA) output breaker.

Condition Prior to Event

Unit(s): 1

Date: March 25, 2024

Time: 04:04 CDT

Reactor Mode(s): 1

Mode(s) Name: Power Operation

Power Level: 100 percent

Description

On March 25, 2024, the main control room (MCR) received a "B" RPS half scram due to a loss of power to the Unit 1 "B" (1B) RPS bus. Operations entered the applicable abnormal operation procedures for the loss of the 1B RPS bus. Field operators determined that the 1B RPS motor-generator (MG) set EPMA output breaker had tripped. The loss of power to the 1B RPS bus resulted in a half-scram condition and closure of isolation valves in multiple systems. The condition was considered an invalid actuation because it was not the result of a valid input signal to the RPS and was not an intentional manual action. Troubleshooting identified the frequency supplied by the MG set was approximately 60 Hz, as expected. All actuations and isolations performed as designed.

The loss of the 1B RPS bus resulted in isolations of the reactor water cleanup system, main steam line drains, reactor coolant system sampling, instrument nitrogen system, reactor recirculation hydraulic power units, and containment monitoring system. The loss of the 1B RPS bus also resulted in the loss of power to portions of the average power range monitor, local power range monitor, oscillation power range monitor, and control rod block systems; as well as portions of the radiation monitoring for the main steam lines, reactor building vent, and fuel pool vent. Appropriate actuations and primary containment isolations occurred as designed.

After the loss of power to the 1B RPS bus, the applicable Technical Specification (TS) timeclocks and actions were entered. Power was restored to the 1B bus by transferring the bus to the alternate power supply in accordance with plant procedures. The half-scram and isolations were reset, and the equipment was returned to operable status. All applicable TS timeclocks and actions were exited.

Cause

The 1B RPS MG set EPMA output breaker tripped due to an underfrequency trip signal from the EPMA trip card. The EPMA trip card was out of range high for the underfrequency setpoint which caused the 1B RPS MG set EPMA output breaker to trip due to underfrequency, even if there was no underfrequency condition present.

Reportability and Safety Analysis

The loss of a RPS bus and associated RPS actuation is considered an invalid actuation because it was not the result of a valid signal and not an intentional manual action. TS Surveillance Requirement (SR) 3.3.8.2.2 requires that the EPMA to have a frequency setpoint of greater than or equal to 57.3 Hz. The EPMA trip card underfrequency setpoint was discovered to be out of range high, therefore the EPMA output breaker trip is considered conservative and in compliance with SR 3.3.8.2.2. The loss of the RPS bus resulted in general containment isolation signals that actuated containment isolation valves in more than one system or multiple main steam isolation valves while the unit was critical. Therefore, the loss of the RPS bus is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

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	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 00

NARRATIVE**Safety System Functional Failure Review**

The system equipment response did not result in a safety system functional failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline." The RPS logic safety function was satisfied by its actuation for designed inputs.

Corrective Actions

Corrective actions taken in response to the conditions were:

- Declared the affected systems inoperable and entered required TS timeclocks and actions.
- Transferred the 1B RPS bus to the alternate power supply.
- Reset the half-scam and primary containment isolations and exited applicable TS timeclocks and actions.
- Replaced trip card due to drifting out of tolerance during attempted recalibration.
- Transferred the 1B RPS bus back to the normal power supply.
- Reviewed the extent of condition for the other Unit 1 RPS EPMA trip cards. No extent of condition was identified for the Unit 1A RPS bus or for the Unit 2 RPS.
- Sent the failed EPMA trip card offsite for failure analysis.

Previous Occurrences**Unit 2 LER 374-2021-002-00**

On May 31, 2021, LaSalle Unit 2 was operating at 100 percent power, when the 2A reactor protection system (RPS) motor generator (MG) set output breaker tripped due to an overload condition, which resulted in a loss of 2A RPS bus. Loss of power to this bus created a half-scam and initiated closure of outboard isolation valves in multiple systems. The condition was an invalid actuation because it was not the result of actual plant conditions and not an intentional manual action. The cause of the issue was a degraded voltage adjustment potentiometer (device 1P) on the 2A RPS MG set, which was causing intermittent resistance readings. The intermittent resistance caused the output voltage of the RPS MG set to fluctuate. As a result, the degraded potentiometer and the voltage regulator were replaced as part of the troubleshooting efforts, and the system was restored back to its original configuration.

Component Failure Data

Manufacturer: General Electric

Device: Unit 1B Electrical Protection Assembly Logic Circuit Board

Component ID: 1C71-S003B, 1B RPS MG Set Electric Power Monitoring Assembly Breaker B (Cat. ID 23326)